



capabilities, including a spectrum analyser with modulation analyser. "Such instruments with power meters cover the complete life cycle of base-station and mobile-station testing from development to maintenance. Research indicates that around 62 per cent of the spectrum analyser market revenue comes from the RF segment. It is anticipated that, with evolving end-user technologies, most spectrum analysers will be RF-compatible in the future. Technologies like 3G/4G deployments, increasing complexity of mobile handsets and Long-Term Evolution (LTE) development efforts are expected to drive the growth in the RF spectrum analyser market in the foreseeable future," explains Thakare.

Want to build a Wi-Fi network that has a longer range, while using less power than existing Wi-Fi? You can try out IEEE 802.11ah wireless networking protocol to do so.

Vishal says that software like Keysight X-series measurement applications let you add on necessary capabilities to support test equipment.

In addition to this, newer standards such as Data Over Cable Service Interface Specification 3.1 (DOCSIS) and 5G analysis are now being supported, too. Newer equipment also adds support for custom in-phase quadrature (IQ) modulation analysis, which supports engineers working on proprietary standards that are non-standard.

Madhukar Tripathi, senior manager, marketing and channel sales at Anritsu India Pvt Ltd, explains that Common Public Radio Interface (CPRI) measurement will be required since LTE deployment is in full swing across the globe. A key need is the ability to analyse the uplink RF spectrum from ground level via CPRI optical-fibre

Major contributors to this report

- **Chandmal Goliya**, director, Kusam Electrical Industries Ltd
- **Madhukar Tripathi**, senior manager, marketing and channel sales, Anritsu India Pvt Ltd
- **Naveen Sharma**, technical manager, Scientific Mes-Technik Pvt Ltd
- **Rufus Danesh**, project engineer, Analog Arts
- **Satish Thakare**, chief technology officer, Sciencetech Technologies Pvt Ltd
- **Vishal Gupta**, senior applications specialist (RF/MW), Keysight Technologies

link, as it can reduce operational cost and increase response time to resolve interference issues. This eliminates the need to call a tower crew to climb to the top of the antenna, and connect a spectrum analyser to the RF sniffer port to look for interference.

"Our MS2830A spectrum analyser is now capable of supporting development and production of equipment for both E-band (70GHz/80GHz) wireless back-haul as well as automobile collision avoidance radar (77GHz/79GHz)," explains Tripathi.

"We also say the launch of two new electromagnetic field (EMF) measurement antennae to the handheld spectrum analyser family like MS2720T, LMR Master S412E is useful for regulatory authorities, monitoring agencies, survey agencies and mobile operators," he adds.

Danesh feels, since the majority of applications for signal analysers—whether these are spectrum analysers, oscilloscopes or voltmeters—are within a laboratory environment, we have still some way to go before we have a true standard set of specification and support for wireless devices that is universally agreed upon. Other new features also lack standardisation in the true sense.

Professional versus educational

There used to be important differences between the features of instruments intended for professional applications and those for

other markets. These days, however, the gap is closing quickly. Low-cost instruments enjoy similar functionalities as the more expensive equipment. Danesh strongly believes that we will soon see the performance of a high-end signal analyser in a device at a fraction of the cost.

He adds, "One area for signal analysis that is becoming popular, particularly for hobbyists, is the application of the instruments to analyse wireless satellite communications. Software support, available with the newest devices, makes this application both fun and easy."

"For the education market, we have launched simple instruments to make the frequency characteristic analysis for students, who can use this analyser to check network parameters in low-frequency ranges, for example, amplitude-frequency, phase-frequency, frequency discrimination and S11, S12 parameters of network. The function is similar to a simple network analyser," explains Goliya.

The overall trend

In recent years, specifications such as integration, automation, portability, remote control and ease of use are becoming important features of a spectrum analyser. "We believe this trend will continue for the foreseeable future," says Danesh.

Sharma feels that the next breakthrough could come in the form of usage of poly-phase filters. "These filters were not possible until today," he adds. **EFY**